INTERNATIONAL GCSE MATHEMATICS A · LINEAR 4MA1 HIGHER

The Night-Before Revision Sheet

One page · one idea · one example

The 16 ideas that come up in every 4MA1 Higher paper. Read the key words, follow the example, then cover it and do it again.



DR ESLAM AHMED
eliteigcse.com

100	2	16	1
MARKS · ONE PAPER	PAPERS · 2 H EACH	IDEAS · 16 PAGES	EXAMPLE PER PAGE

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BOTTOM LINE Write every stage. Store numbers in the calculator and round *once*, at the end, to what the question asks (3 s.f. or 1 d.p.). Copy the final answer onto the answer line.

1 Solving with a Graph — the Straight Line

Make the left side the graph you drew; whatever is left on the right is the line you draw.

WHEN YOU SEE → YOU DO

"Use your graph to solve $f(x) = 0$"	→ read where the curve crosses the x-axis ($y = 0$)
"Use your graph to solve $f(x) = k$"	→ draw the horizontal line $y = k$
"By drawing a suitable straight line, solve …"	→ rearrange: graph's equation = line, draw the line, read x

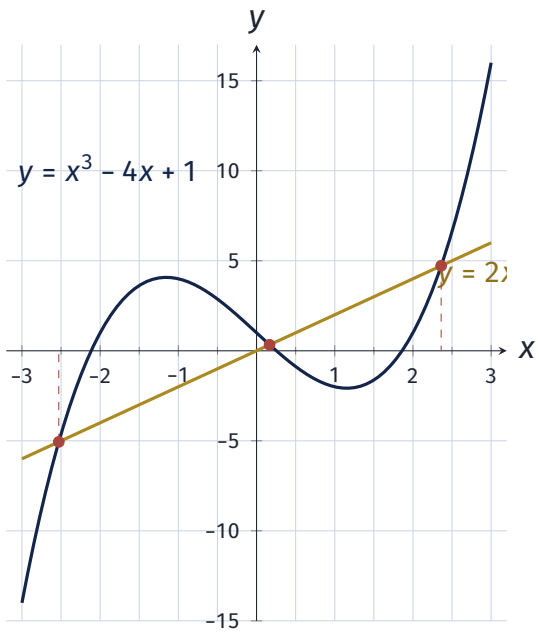
EXAMPLE

The graph of $y = x^3 - 4x + 1$ is drawn for $-3 \le x \le 3$.
By drawing a suitable straight line on the grid, solve $x^3 - 6x + 1 = 0$.

WORKED SOLUTION

1. **WRITE THE TWO EQUATIONS**
   Graph: $x^3 - 4x + 1$
   To solve: $x^3 - 6x + 1 = 0$
2. **MAKE THE LEFT SIDE THE GRAPH**
   Add $2x$ to both sides:
   $x^3 - 4x + 1 = 2x$
3. **THE RIGHT SIDE IS THE LINE**
   Draw $y = 2x$: points $(-3, -6)$, $(0, 0)$, $(3, 6)$ — use a ruler.
4. **READ THE CROSSINGS**
   Go down from each crossing to the x-axis.

ANSWER $x \approx -2.5,\ 0.2,\ 2.4$



TRAP The answer is the x-values only, not the points. Do not solve with the formula — the mark is for drawing the line. A line like $y = 2x$ goes through the origin; $y = 2x - 1$ cuts the y-axis at -1.

2 Histograms

Area is frequency. Height is frequency density.

WHEN YOU SEE → YOU DO

"Draw a histogram"	→ frequency density = frequency ÷ class width
Histogram given, "how many …"	→ frequency = FD × width (area of the bar)
"More than 25" but the class is 20–30	→ take the part of the bar: (30 – 25)× FD

EXAMPLE

The table gives the weights, w kg, of 105 parcels. (a) Draw a histogram. (b) Work out an estimate for the number of parcels heavier than 25 kg.

Weight	$0 < w \le 10$	$10 < w \le 15$	$15 < w \le 20$	$20 < w \le 30$	$30 < w \le 50$
Frequency	15	20	32	28	10

WORKED SOLUTION

1 FREQUENCY DENSITY FOR EACH CLASS

15 ÷ 10 = 1.5 20 ÷ 5 = 4

32 ÷ 5 = 6.4

28 ÷ 10 = 2.8 10 ÷ 20 = 0.5

2 DRAW THE BARS

Widths are the classes, heights are the FD. No gaps. Label the axis *frequency density*.

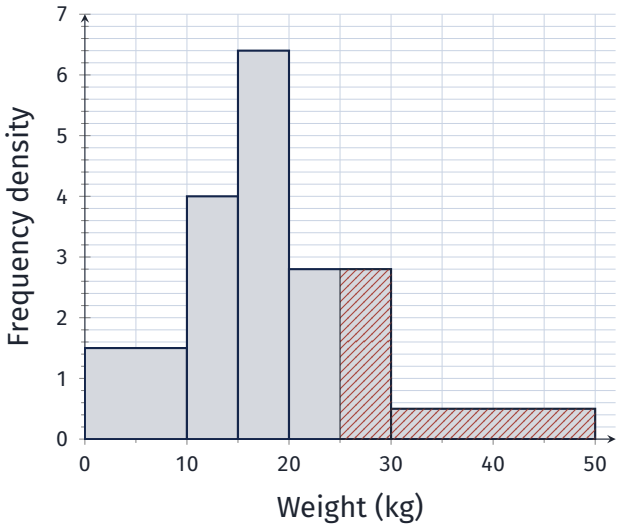
3 HEAVIER THAN 25 KG

Part of the 20–30 bar:

(30 – 25) × 2.8 = 14

Whole 30–50 bar: 10

ANSWER (b) 14 + 10 = 24 parcels



TRAP Never plot frequency as the height. If the vertical axis has no numbers, use one bar you know: frequency ÷ number of squares gives what one square is worth.

3 Tree Diagrams — Without Replacement

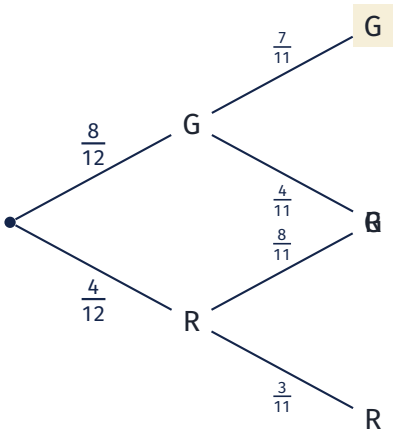
The second set of branches has one less on the bottom — and one less of the colour taken.

WHEN YOU SEE → YOU DO

"takes … then takes another", "does not replace", "eats"	→ second branches: total −1
"both the same colour" / "different colours"	→ add the paths: *GG* + *RR* or *GR* + *RG*
"at least one"	→ 1 − *P*(none)

EXAMPLE

A bag has 8 green and 4 red sweets. Ali takes a sweet at random and eats it. Then he takes a second sweet at random. Work out the probability that he takes at least one red sweet.



WORKED SOLUTION

1. **FILL THE TREE**
   First: $\frac{8}{12}$, $\frac{4}{12}$. Second: out of 11, and the colour taken goes down by 1.
2. **"AT LEAST ONE RED" = NOT GG**
   $P(GG) = \frac{8}{12} \times \frac{7}{11} = \frac{56}{132}$
3. **SUBTRACT FROM 1**
   $1 - \frac{56}{132} = \frac{76}{132}$

ANSWER $\frac{19}{33}$

TRAP Multiply along a path, add between paths. "Different colours" needs BOTH orders: *GR* and *RG*. Three sweets? Same idea: $\frac{a}{n} \times \frac{b}{n-1} \times \frac{c}{n-2}$ and count the orders.

4 Algebraic Fractions

Factorise everything first. Then cancel, or find a common denominator.

WHEN YOU SEE → YOU DO

"Simplify fully" a fraction	→	factorise top and bottom, cancel whole brackets
"Express as a single fraction", + or −	→	common denominator = product of the different brackets
÷ between two fractions	→	flip the second and multiply

EXAMPLE

(a) Simplify fully $\dfrac{3x^2 - 10x - 8}{x^2 - 16}$ simplest form.

(b) Express $\dfrac{2}{x+3} + \dfrac{5}{x-2}$ as a single fraction in its

WORKED SOLUTION

1 (A) FACTORISE TOP AND BOTTOM

$3x^2 - 10x - 8 = (3x + 2)(x - 4)$ $x^2 - 16 = (x - 4)(x + 4)$ (difference of two squares)

2 (A) CANCEL THE COMMON BRACKET

$$\frac{(3x+2)\cancel{(x-4)}}{\cancel{(x-4)}(x+4)} = \frac{3x+2}{x+4}$$

3 (B) COMMON DENOMINATOR

$$\frac{2(x-2)+5(x+3)}{(x+3)(x-2)} = \frac{2x-4+5x+15}{(x+3)(x-2)}$$

ANSWER (a) $\dfrac{3x+2}{x+4}$ (b) $\dfrac{7x+11}{(x+3)(x-2)}$

TRAP You can only cancel a whole bracket that multiplies — never cancel the x from $\frac{x+3}{x}$. With a minus in front of a fraction, bracket the whole top: $-3(x+1) = -3x - 3$.

5 Bounds in a Calculation

To make an answer as big as possible: big on top, small on the bottom.

WHEN YOU SEE → YOU DO

"correct to the nearest 5"	→ half of 5 each side: ±2.5
"correct to 1 decimal place"	→ ±0.05
Upper bound of $\frac{a}{b}$ or $a - b$	→ UB of a with LB of b
"to a suitable degree of accuracy"	→ find UB and LB, round both; keep what they agree on

EXAMPLE

A runner ran 86 m, correct to the nearest metre, in 9.4 seconds, correct to 1 decimal place. By considering bounds, work out the average speed to a suitable degree of accuracy. Give a reason.

WORKED SOLUTION

1. **BOUNDS OF EACH VALUE**
   distance: $85.5 \le d < 86.5$ time: $9.35 \le t < 9.45$
2. **UPPER BOUND OF SPEED**
   $\frac{\text{UB } d}{\text{LB } t} = \frac{86.5}{9.35} = 9.2513\ldots$
3. **LOWER BOUND OF SPEED**
   $\frac{\text{LB } d}{\text{UB } t} = \frac{85.5}{9.45} = 9.0476\ldots$
4. **COMPARE**
   to 2 s.f.: 9.3 and 9.0 — different. To 1 s.f.: 9 and 9 — the same.

ANSWER 9 m/s, because the upper and lower bounds both round to 9 to 1 significant figure.

TRAP Nearest 10 → ±5; nearest 0.1 → ±0.05. The reason is part of the mark — write it.

6 Surds

Expand like brackets; to rationalise, multiply top and bottom by the conjugate.

WHEN YOU SEE → YOU DO

$\frac{k}{\sqrt{a}}$	→ multiply by $\frac{\sqrt{a}}{\sqrt{a}}$
$\frac{k}{a-\sqrt{b}}$	→ multiply by $\frac{a+\sqrt{b}}{a+\sqrt{b}}$ (change the sign)
"in the form $c+d\sqrt{b}$ where c, d are integers"	→ rationalise, then simplify every part

EXAMPLE

(a) Expand and simplify $(2+\sqrt{5})(3-\sqrt{5})$ (b) Show that $\frac{6}{3-\sqrt{3}}$ can be written as $c+\sqrt{3}$ where c is an integer.

WORKED SOLUTION

1. **(A) FOUR TERMS**
$6-2\sqrt{5}+3\sqrt{5}-5=1+\sqrt{5}$
2. **(B) MULTIPLY BY THE CONJUGATE**
$\frac{6}{3-\sqrt{3}}\times\frac{3+\sqrt{3}}{3+\sqrt{3}}=\frac{6(3+\sqrt{3})}{9-3}$
3. **(B) SIMPLIFY**
$\frac{6(3+\sqrt{3})}{6}=3+\sqrt{3}$

ANSWER (a) $1+\sqrt{5}$ (b) $3+\sqrt{3}$, so $c=3$

TRAP $\sqrt{5}\times\sqrt{5}=5$, not 25. The bottom $(3-\sqrt{3})(3+\sqrt{3})=9-3$: the middle terms cancel. Useful: $\sqrt{12}=2\sqrt{3}$, $\sqrt{50}=5\sqrt{2}$.

7 Completing the Square

Take out the x^2 number, halve the x number, then tidy the constant.

WHEN YOU SEE → YOU DO

$(x + p)^2 + q$	→ p = half the x coefficient; q = constant $-p^2$
$a(x + p)^2 + q$	→ factor a out of the x^2 and x terms first
"in the form $a - b(x - c)^2$" ($-x^2$)	→ take out the negative number
"Hence solve", "give answers in surd form"	→ $(x + p)^2 = -q$, square root, ±

EXAMPLE

Express $5 + 8x - 2x^2$ in the form $a - b(x - c)^2$ where a, b and c are integers.

WORKED SOLUTION

1. **TAKE OUT -2 FROM THE x TERMS**
   $-2x^2 + 8x + 5 = -2(x^2 - 4x) + 5$
2. **COMPLETE THE SQUARE INSIDE**
   $x^2 - 4x = (x - 2)^2 - 4$
3. **EXPAND THE OUTSIDE AND TIDY**
   $-2\left[(x - 2)^2 - 4\right] + 5 = -2(x - 2)^2 + 8 + 5$

ANSWER $13 - 2(x - 2)^2$: $a = 13$, $b = 2$, $c = 2$

HENCE SOLVE — THE SAME IDEA

$x^2 - 6x + 2 = 0 \Rightarrow (x - 3)^2 - 9 + 2 = 0 \Rightarrow (x - 3)^2 = 7 \Rightarrow x = 3 \pm \sqrt{7}$

TRAP When the -2 multiplies back, the -4 becomes $+8$. Check by expanding your answer — it must give the question back.

8 Sine Rule, Cosine Rule, Area

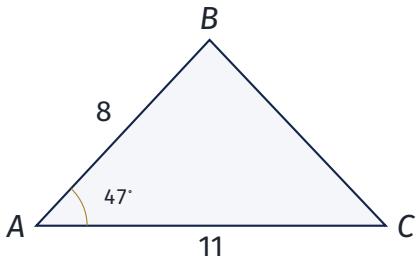
Label the sides with small letters opposite the big-letter angles — then choose.

WHEN YOU SEE → YOU DO

Two sides and the angle BETWEEN them	→ cosine rule for the third side; area = $\frac{1}{2}ab\sin C$
Three sides, find an angle	→ $\cos A = \dfrac{b^2 + c^2 - a^2}{2bc}$
A side and its opposite angle are known	→ sine rule $\dfrac{a}{\sin A} = \dfrac{b}{\sin B}$

EXAMPLE

In triangle ABC, $AB = 8$ cm, $AC = 11$ cm and angle $BAC = 47^\circ$.
(a) Work out the length of BC. (b) Work out the area of the triangle. Give answers to 3 s.f.



WORKED SOLUTION

1. **TWO SIDES + ANGLE BETWEEN → COSINE RULE**
   $BC^2 = 8^2 + 11^2 - 2(8)(11)\cos 47^\circ = 64.97\ldots$
2. **SQUARE ROOT**
   $BC = 8.0602\ldots$
3. **AREA**
   $\frac{1}{2}(8)(11)\sin 47^\circ = 32.179\ldots$

ANSWER (a) 8.06 cm (b) 32.2 cm^2

TRAP Calculator in DEGREES. Square root at the end of the cosine rule. When the angle is obtuse, cos is negative — the "$-2bc\cos A$" becomes plus.

9 3D Trigonometry — Line and Plane

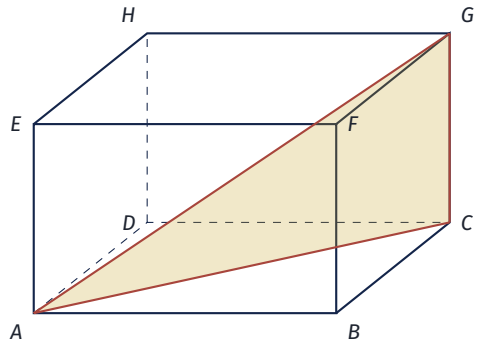
Drop a perpendicular from the top of the line to the plane; the angle sits in a right-angled triangle.

WHEN YOU SEE → YOU DO

"angle between the line *AG* and the plane *ABCD*"	→ triangle *A*–*C*–*G*: *C* is directly below *G*
Diagonal of the base rectangle	→ Pythagoras: $\sqrt{l^2 + w^2}$
"angle of elevation of *G* from *D*"	→ the angle at *D* between the horizontal and *DG*

EXAMPLE

ABCDEFGH is a cuboid with *AB* = 8 cm, *BC* = 6 cm and *CG* = 5 cm.
Work out the size of the angle between *AG* and the plane *ABCD*. Give your answer to 1 d.p.



WORKED SOLUTION

1. **FIND THE BASE DIAGONAL *AC***
   $AC = \sqrt{8^2 + 6^2} = 10$
2. **RIGHT ANGLE AT *C***
   opposite $CG = 5$, adjacent $AC = 10$
3. **TAN**
   $\tan\theta = \frac{5}{10} \Rightarrow \theta = 26.565\ldots^\circ$

ANSWER 26.6°

TRAP The angle with the plane is NOT the angle with an edge. Draw the right-angled triangle separately and label it before any trig.

10 Circle Theorems — With Reasons

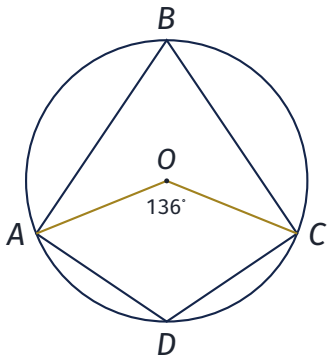
Find one angle at a time and write the theorem next to it.

WHEN YOU SEE → YOU DO

Angle at the centre and at the circumference	→ centre = 2× circumference
Four points on the circle (cyclic quadrilateral)	→ opposite angles add to 180°
Tangent and a chord	→ alternate segment: equals the angle in the other segment
Tangent and a radius	→ meet at 90°

EXAMPLE

A, *B*, *C* and *D* are points on a circle, centre *O*. Angle *AOC* = 136°.
Work out (a) angle *ABC* (b) angle *ADC*. Give a reason for each step.



WORKED SOLUTION

1. **(A) CENTRE IS DOUBLE THE CIRCUMFERENCE**
   ABC = 136 ÷ 2 = 68° — *the angle at the centre is twice the angle at the circumference.*
2. **(B) CYCLIC QUADRILATERAL**
   ADC = 180 − 68 = 112° — *opposite angles of a cyclic quadrilateral add up to* 180°.

ANSWER (a) 68° (b) 112°

TRAP Write the reason in words, not only the number — the mark needs it. Check which arc: *D* is on the same side as the 136° angle, so it is the obtuse one.

11 Vectors — a Route and a Proof

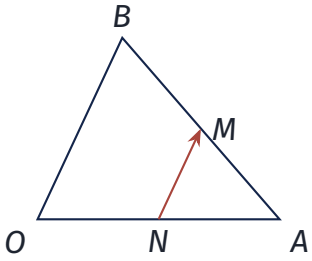
Write the route as letters first: $\vec{NM} = \vec{NO} + \vec{OM}$. Then swap in **a** and **b**.

WHEN YOU SEE → YOU DO

$\vec{AB}$ **from** $\vec{OA}$ **and** $\vec{OB}$	$\longrightarrow$ $\vec{AB} = \mathbf{b} - \mathbf{a}$
Midpoint M **of** AB	$\longrightarrow$ $\vec{OM} = \frac{1}{2}(\mathbf{a} + \mathbf{b})$
"Show that ... is parallel to ..."	$\longrightarrow$ one vector is a number × the other
"Show that P, Q, R **lie on a straight line"**	$\longrightarrow$ $\vec{PR} = k\vec{PQ}$ AND a common point

EXAMPLE

OAB is a triangle. $\vec{OA} = \mathbf{a}$ and $\vec{OB} = \mathbf{b}$. M is the midpoint of AB and N is the midpoint of OA.
Show that NM is parallel to OB.



WORKED SOLUTION

1. **ROUTE THROUGH** O
$\vec{NM} = \vec{NO} + \vec{OM}$
2. **EACH PIECE**
$\vec{NO} = -\frac{1}{2}\mathbf{a}$, $\vec{OM} = \frac{1}{2}(\mathbf{a} + \mathbf{b})$
3. **ADD**
$\vec{NM} = -\frac{1}{2}\mathbf{a} + \frac{1}{2}\mathbf{a} + \frac{1}{2}\mathbf{b} = \frac{1}{2}\mathbf{b}$

ANSWER $\vec{NM} = \frac{1}{2}\vec{OB}$, so NM is parallel to OB (and half its length).

TRAP Going against an arrow changes the sign. The conclusion sentence is a mark: say "a multiple of **b**, so parallel".

12 Functions — Composite and Inverse

$fg(x)$ means do g FIRST, then f. The inverse undoes the function.

WHEN YOU SEE → YOU DO

$fg(2)$	⟶	work out $g(2)$, then put the answer into f
$gf(x)$	⟶	replace every x in g with the whole of $f(x)$
$f^{-1}(x)$	⟶	$y = f(x)$, swap to make x the subject, rename
"which value must be excluded"	⟶	the value that makes a denominator 0

EXAMPLE

$f(x) = 3x - 2$ $\quad g(x) = \dfrac{1}{x+1}$

Find (a) $fg(2)$ (b) $gf(x)$ (c) $f^{-1}(x)$ (d) $g^{-1}(x)$

WORKED SOLUTION

1. **(A) g FIRST**
   $g(2) = \frac{1}{3}$, then $f\left(\frac{1}{3}\right) = 1 - 2 = -1$
2. **(B) PUT f INSIDE g**
   $$gf(x) = \frac{1}{(3x-2)+1} = \frac{1}{3x-1}$$
3. **(C) REARRANGE $y = 3x - 2$**
   $$x = \frac{y+2}{3}$$
4. **(D) REARRANGE $y = \dfrac{1}{x+1}$**
   $$x + 1 = \frac{1}{y} \Rightarrow x = \frac{1}{y} - 1$$

ANSWER (a) -1 (b) $\dfrac{1}{3x-1}$ (c) $f^{-1}(x) = \dfrac{x+2}{3}$ (d) $g^{-1}(x) = \dfrac{1-x}{x}$

TRAP $fg(x)$ is NOT $f(x) \times g(x)$. In the final answer of an inverse, write it in terms of x, not y.

13 Differentiation — Turning Points

Differentiate, set the gradient to zero, solve, then find y.

WHEN YOU SEE → YOU DO

$y = ax^n$	$\longrightarrow \frac{dy}{dx} = nax^{n-1}$ (a number on its own goes)
"turning point", "maximum", "minimum"	$\longrightarrow \frac{dy}{dx} = 0$
"gradient at $x = 2$"	$\longrightarrow$ substitute into $\frac{dy}{dx}$, not into y
"maximum or minimum?"	$\longrightarrow$ look at the shape, or the gradient just before and after

EXAMPLE

Find the coordinates of the turning points of $y = x^3 - 6x^2 + 9x + 1$ and say which is a maximum.

WORKED SOLUTION

1. **DIFFERENTIATE**
$\frac{dy}{dx} = 3x^2 - 12x + 9$
2. **SET EQUAL TO 0**
$3(x^2 - 4x + 3) = 0 \Rightarrow 3(x - 1)(x - 3) = 0 \Rightarrow x = 1$ or $x = 3$
3. **FIND y**
$x = 1$: $y = 1 - 6 + 9 + 1 = 5$ $\quad$ $x = 3$: $y = 27 - 54 + 27 + 1 = 1$
4. **WHICH IS WHICH**
A positive x^3 curve goes up, down, up: the first turning point is the maximum.

ANSWER maximum (1, 5), minimum (3, 1)

TRAP Give BOTH coordinates. The gradient is the $\frac{dy}{dx}$ value; the point is the (x, y) pair.

14 Simultaneous Equations — Line and Curve

Make y (or x) the subject of the LINE and substitute into the curve.

WHEN YOU SEE → YOU DO

One linear, one with x^2 or y^2	→ substitute the line into the curve
$x^2 + y^2 = r^2$	→ a circle, centre O, radius r
Quadratic in x appears	→ solve, then find each y from the LINE

EXAMPLE

Solve the simultaneous equations $y = 2x - 1$ $x^2 + y^2 = 13$

WORKED SOLUTION

1. **SUBSTITUTE**
   $x^2 + (2x - 1)^2 = 13$
2. **EXPAND AND COLLECT**
   $x^2 + 4x^2 - 4x + 1 = 13 \Rightarrow 5x^2 - 4x - 12 = 0$
3. **FACTORISE**
   $(5x + 6)(x - 2) = 0 \Rightarrow x = 2$ or $x = -\frac{6}{5}$
4. **y FROM THE LINE**
   $x = 2$: $y = 3$ $x = -\frac{6}{5}$: $y = -\frac{12}{5} - 1 = -\frac{17}{5}$

ANSWER $x = 2,\ y = 3$ or $x = -1.2,\ y = -3.4$

TRAP $(2x - 1)^2$ has a middle term: $4x^2 - 4x + 1$. Pair the answers: each x goes with its own y.

15 Cumulative Frequency — Median and IQR

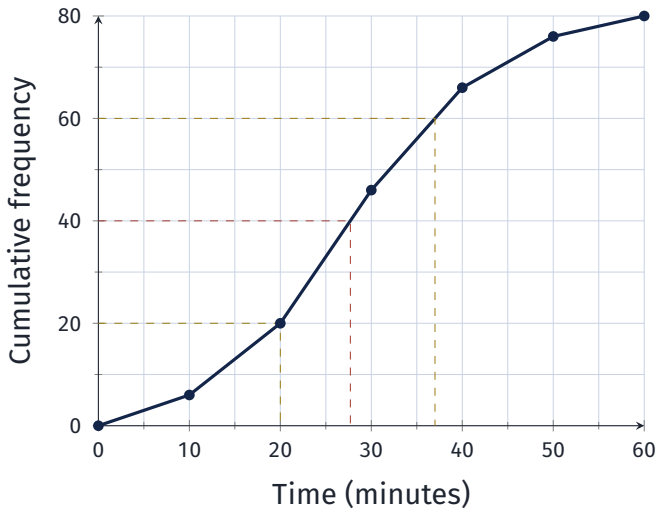
Read across from the cumulative frequency, then down to the value.

WHEN YOU SEE → YOU DO

Median (n people)	→	read at $\frac{n}{2}$
Lower / upper quartile	→	read at $\frac{n}{4}$ and $\frac{3n}{4}$; IQR = UQ − LQ
"how many more than 45"	→	read up from 45, then total − that reading

EXAMPLE

The graph shows the times, in minutes, that 80 people waited. Find estimates for the median and the interquartile range.



WORKED SOLUTION

1. **MEDIAN AT 40**
   read across from 40: about 28 minutes
2. **QUARTILES AT 20 AND 60**
   LQ = 20, UQ ≈ 37
3. **IQR**
   37 − 20 = 17

ANSWER median ≈ 28 min, IQR ≈ 17 min

TRAP Plot cumulative frequency at the END of each class. Read the quartiles at 20 and 60 — not at 25% of the time axis.

16 Similar Shapes — Area and Volume

Find the LENGTH scale factor first; then square it for area, cube it for volume.

WHEN YOU SEE → YOU DO

Length scale factor k	⟶	area $\times k^2$, volume $\times k^3$
Areas given	⟶	$k = \sqrt{\text{area ratio}}$
Volumes given	⟶	$k = \sqrt[3]{\text{volume ratio}}$

EXAMPLE

Two solid cones are mathematically similar. Their surface areas are 45 cm^2 and 180 cm^2. The volume of the smaller cone is 30 cm^3. Work out the volume of the larger cone.

WORKED SOLUTION

1. **AREA SCALE FACTOR**
   $180 \div 45 = 4$
2. **LENGTH SCALE FACTOR**
   $\sqrt{4} = 2$
3. **VOLUME SCALE FACTOR**
   $2^3 = 8$

ANSWER $30 \times 8 = 240$ cm^3

TRAP Do not multiply the volume by 4 (the area factor). Always go back to the length factor first.